Entities and Attributes

The ER diagram consists of five main entities:

1. **Employee**
   - `employeeID`: **Primary Key**
   - `email`
   - `password`
2. **Department**
   - `departmentID`: **Primary Key**
   - `departmentName`
3. **Payroll**
   - `payrollID`: **Primary Key**
   - `basicSalary`
   - `bonus`
   - `deductions`
   - `netSalary`
   - `employeeID`: Foreign Key (references Employee)
4. **Project**
   - `projectID`: **Primary Key**
   - `projectName`
   - `startDate`
   - `endDate`
   - `departmentID`: Foreign Key (references Department)
5. **Attendance**
   - `attendanceID`: **Primary Key**
   - `date`
   - `status`
   - `employeeID`: Foreign Key (references Employee)

Relationships and Cardinality

The relationships between these entities are as follows:

- **Department and Employee (One-to-Many)**
  - A single `Department` can have one or more `Employees`.
  - Each `Employee` must belong to exactly one `Department`.
- **Employee and Payroll (One-to-Many)**
  - A single `Employee` can have zero or more `Payroll` records (e.g., a record for each salary payment).
  - Each `Payroll` record is associated with exactly one `Employee`.
- **Department and Project (One-to-Many)**
  - A single `Department` can manage zero or more `Projects`.
  - Each `Project` is managed by exactly one `Department`.

- **Employee and Attendance (One-to-Many)**
  - A single `Employee` can have many `Attendance` records.
  - Each `Attendance` record belongs to exactly one `Employee`.
- **Employee and Project (Many-to-Many)**
  - An `Employee` can be assigned to one or more `Projects`.
  - A `Project` can have zero or more `Employees` working on it.
  - In an ER diagram, this many-to-many relationship is resolved by creating an **associative entity** (or junction table), such as `Employee_Project`, which would contain `employeeID` and `projectID` as foreign keys.